

Foundation (Jalapeno)

Q1. Solve for x : $2^x = 8$

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$
- (E) $x = 6$

Q2. If $3^x = 27$, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q3. In a sports tournament, the number of participants doubles every round. If there are 2^5 participants initially, how many will there be after 3 rounds?

- (A) 2^8
- (B) 2^{10}
- (C) 2^{12}
- (D) 2^{15}
- (E) 2^{20}

Q4. A basketball player scores 2^x points in a game. If she scores 16 points, what is x ?

- (A) $x = 2$
- (B) $x = 3$
- (C) $x = 4$
- (D) $x = 5$

(E) $x = 6$

Q5. Solve for x : $4^x = 64$

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q6. A cyclist travels 3^x miles in a day. If she travels 27 miles, what is x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q7. Solve for x : $2^x = 32$

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

Q8. If $x^2 = 2^4$, what is x ?

- (A) $x = 2$
- (B) $x = 4$
- (C) $x = 6$
- (D) $x = 8$
- (E) $x = 10$

Open Ended Questions (FRQ)

Q9. Solve for x : $5^x = 125$. Show your work and explain your reasoning.

Q10. A runner completes a 2^x mile course in a day. If she completes 8 miles, what is x ? Show your calculation steps.

Chef's Tips

- Ensure students use the correct order of operations
- Encourage students to check their work
- Remind students to show their work for free response questions